A Lambert–W and Hopf Atlas for a Scalar Retarded Semigroup

Route-A dynamics study — HCS-C210

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Revision status. Revision checkpoint: all Hopf crossings and retarded root-continuity proof boundary added. AI-use disclosure: language assistance was used for drafting and code review; all displayed identities and receipts are independently checked.

Abstract

We freeze the scalar retarded equation $x'(t) = -ax(t) - bx(t-\tau)$ on $C([-\tau, 0]; \mathbb{C})$ for all nonnegative a, b, τ. One theorem package closes the characteristic roots through Lambert W, the exact method-of-steps fundamental solution, eventual compactness of the history semigroup, algebraic multiplicities, and the complete stability/Hopf boundary. The result is deliberately source-local: a characteristic function is not renamed a Fredholm determinant, and no arithmetic or target spectral claim is made. The strict Route-A record is `(A0_FAIL,A1_FAIL,A2_FAIL,A3_FAIL,A4_FAIL)`.

1 Owner, clock, and scope

For $\tau > 0$ write $X = C([-\tau, 0]; \mathbb{C})$ with the supremum norm and $x_t(\theta) = x(t+\theta)$. The generator is

$$A\phi = \phi', \qquad D(A) = \{\phi \in C^1 : \phi'(0) = -a\phi(0) - b\phi(-\tau)\}.$$

The clock is physical elapsed time and the normalization is $r(0) = 1$, $r(t) = 0$ for $t < 0$. The case $\tau = 0$ is the scalar ODE. These choices are frozen before any calculation; no fitted parameter or target data is used.

2 Characteristic spectrum

Theorem 1 (Lambert branches and multiplicity)**.** *For $\tau > 0$ and $b > 0$, the characteristic function*

$$\Delta(\lambda) = \lambda + a + be^{-\lambda\tau}$$

has precisely the roots

$$\lambda_k = -a + \tau^{-1} W_k(-b\tau e^{a\tau}), \qquad k \in \mathbb{Z},$$

counted with algebraic multiplicity. A multiple root is possible exactly when $b\tau e^{a\tau} = e^{-1}$; it then has multiplicity two and is $\lambda = -a - \tau^{-1}$.

Indeed $We^W = -b\tau e^{a\tau}$ gives the displayed substitution, while $\Delta' = 1 - b\tau e^{-\lambda\tau}$ and $\Delta'' = b\tau^2 e^{-\lambda\tau}$ give the multiplicity assertion. The branch point is not a Hopf boundary in general; the two loci are kept separate in the evidence.

3 Method of steps and compactness

Proposition 1 (Exact fundamental solution)**.** *For $\tau > 0$,*

$$r(t) = \sum_{n=0}^{\lfloor t/\tau \rfloor} \frac{(-b)^n}{n!} e^{-a(t-n\tau)}(t-n\tau)^n, \qquad t \geq 0.$$

At the zero-delay face the fundamental solution is instead $r(0) = 1$ and $r(t) = e^{-(a+b)t}$ for $t > 0$; the delayed formula is not silently continued through $\tau = 0$. The Laplace transform is

$$\frac{1}{\Delta(s)} = \frac{1}{s+a} \sum_{n \geq 0} \frac{(-b)^n e^{-ns\tau}}{(s+a)^n},$$

and inversion term by term gives the formula. The history solution operators $T(t)$ form a strongly continuous semigroup. For $t > \tau$, a bounded set of histories is mapped into a bounded equicontinuous subset of C^1; hence Arzelà–Ascoli makes $T(t)$ compact. The standard eventual-compact spectral mapping statement therefore gives

$$\sigma(T(t)) \setminus \{0\} = \exp\bigl(t\sigma(A)\bigr).$$

At a nonzero semigroup eigenvalue μ, algebraic multiplicity is the sum of the characteristic-root multiplicities over all roots satisfying $e^{t\lambda} = \mu$; exponential collisions are thus aggregated. This is an operator theorem for the delay owner, not a target determinant construction.

4 Stability and Hopf surface

Theorem 2 (All nonnegative parameters)**.** *If $a \geq b$ and $(a,b) \neq (0,0)$, the semigroup is exponentially stable for every finite τ. If $b > a$, set*

$$\omega = \sqrt{b^2 - a^2}, \qquad \tau_c = \frac{\arccos(-a/b)}{\omega}.$$

It is exponentially stable exactly when $0 \leq \tau < \tau_c$; at τ_c the only imaginary roots are the simple pair $\pm i\omega$, and for $\tau > \tau_c$ there is a right-half-plane conjugate pair. The crossing is transversal:

$$\frac{d\,\mathrm{Re}\,\lambda}{d\tau} = \frac{\omega^2}{(1+a\tau)^2 + (\omega\tau)^2} > 0.$$

To see the boundary, substitute $i\omega$ into $\Delta = 0$ to obtain $a + b\cos(\omega\tau) = 0$ and $\omega = b\sin(\omega\tau)$. A crossing can therefore occur only for $b > a$. Continuity from $\tau = 0$ and the positive crossing derivative give the stated regions. More explicitly, all Hopf crossings are at

$$\tau_m = \frac{\arccos(-a/b) + 2\pi m}{\omega}, \qquad m = 0, 1, 2, \ldots,$$

and each pair crosses from left to right; hence stability holds before the first boundary and never returns afterwards. At $a = b > 0$ no finite-delay imaginary root occurs, although the spectral gap closes in the limit $\tau \to \infty$. At $\tau = 0$ one has $x' = -(a+b)x$; at $b = 0$ the delay term is absent; and at $a = b = 0$ histories become constant after one delay. The root-count continuation used here is the standard retarded-semigroup argument of Hale and Verduyn Lunel; the finite sentinel grid is not used to prove this all-delay assertion.

5 Evidence and independent checks

The receipt contains twelve exact rational parameter cases, thirteen quarter time points per case, and three symbolic Hopf controls. The producer writes only canonical rational strings. A separate checker reconstructs all 156 method-of-steps cells, enforces recursive key closure and locks every scope field. A SymPy program independently verifies the Lambert substitution, Δ' criterion, the Gamma/Laplace term and the Hopf modulus. Clean replay is byte-identical; 24 repaired-hash mutations (including two unknown-key attacks) and one stale-hash mutation are rejected.

6 Route-A boundary

The model has no intrinsic rational-prime carrier and no isolated primitive orbit ledger. Its exact Δ is a characteristic function, not a Fredholm determinant or target divisor. No global target analytic structure, same-clock self-adjoint lift, Euler factor, root number, automorphy object or Hilbert–Pólya operator is claimed. Thus the evaluator record is

$$\texttt{(A0\_FAIL,A1\_FAIL,A2\_FAIL,A3\_FAIL,A4\_FAIL)}, \qquad \texttt{ROUTE\_A\_REJECTED},$$

with Route B false and scope literal `NO_BAD_EULER_OR_ROOT_NUMBER`. The internal source theorem is the contribution; the finite ledger is only a reproducibility sentinel.

References

[1] J. K. Hale and S. M. Verduyn Lunel, *Introduction to Functional Differential Equations*, Applied Mathematical Sciences 99, Springer, 1993, doi:10.1007/978-1-4612-4342-7.